

Scalability & Future Plan

Date	3 July 2024
Team ID	XXXXXX
Project Name	A comprehensive measure of well being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only local deployment using the Flask development server	Limits access to a single local machine	High
2	Uses a serialized model (model.pkl) without a database for storing prediction history.	Prediction records are not saved for future analysis.	Medium
3	Accepts only four input parameters for HDI prediction.	Limits flexibility for incorporating additional development indicators	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of local users.	Deploy on a cloud platform with a production-ready web server to support multiple users.
2	Data Storage	Uses only <code>model.pkl</code> ; no database is connected.	Integrate a database such as MySQL or PostgreSQL to store user inputs and prediction history.
3	Performance	Loads a single trained model for predictions.	Optimize the model, enable caching, and improve request handling for faster response times.
4	Security	Basic input handling without authentication.	Add user authentication, HTTPS, and stronger input validation to improve security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application to a cloud platform.	1–2 Months	Enables online access for users
Phase 2	Add user login and prediction history storage.	2-3 Months	Add user login and prediction history storage.
Phase 3	Enhance the machine learning model with additional development indicators.	3-4 Months	Increases prediction accuracy and reliability.
Phase 4	Develop a mobile-friendly interface and interactive analytics dashboard.	4-6 Months	Improves accessibility and data visualization for users.